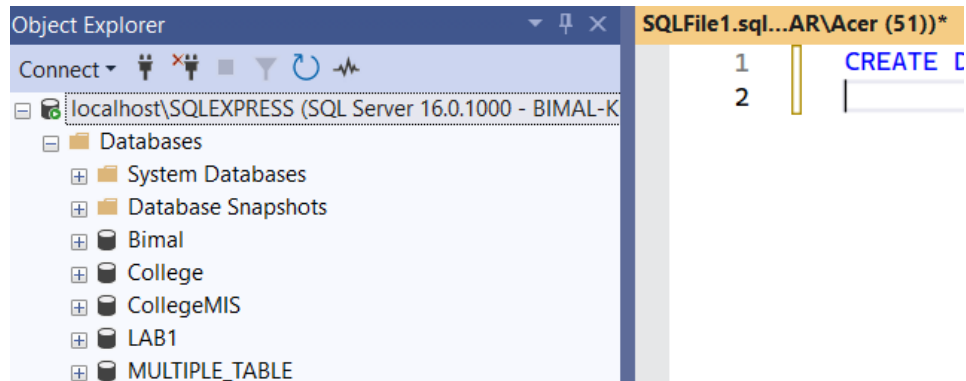


Lab Assignment 2: Use of Multiple tables

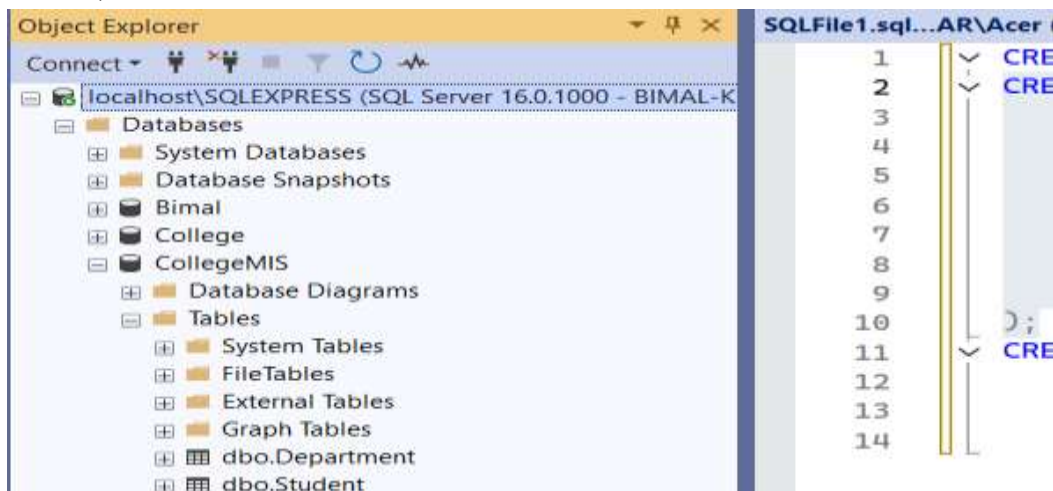
A. Creating Database:

Query: CREATE DATABASE CollegeMIS;



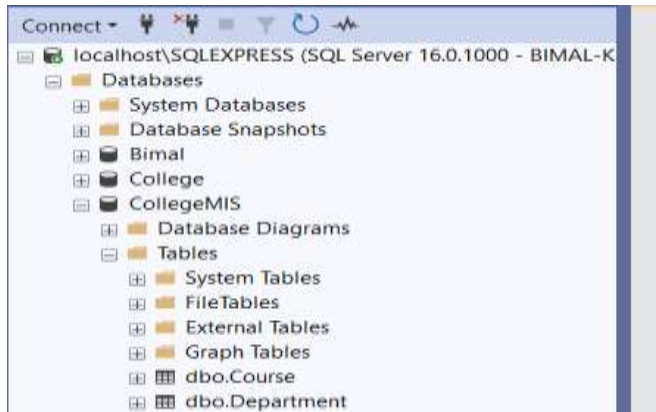
B. Creating Table of Student:

Query: CREATE TABLE Student (
StudentID INT PRIMARY KEY Y,
Name VARCHAR(50),
Gender CHAR(1),
DOB DATE,
DepartmentID INT,
Email VARCHAR(100),
FOREIGN KEY (DepartmentID) REFERENCES Department(DepartmentID)
);



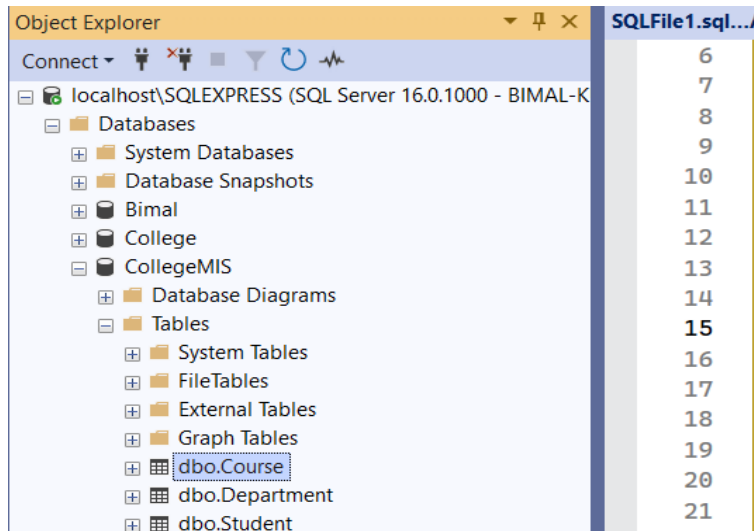
C. Creating Table of Department:

Query: CREATE TABLE Department (
 DepartmentID INT PRIMARY KEY,
 DeptName VARCHAR(50)
);



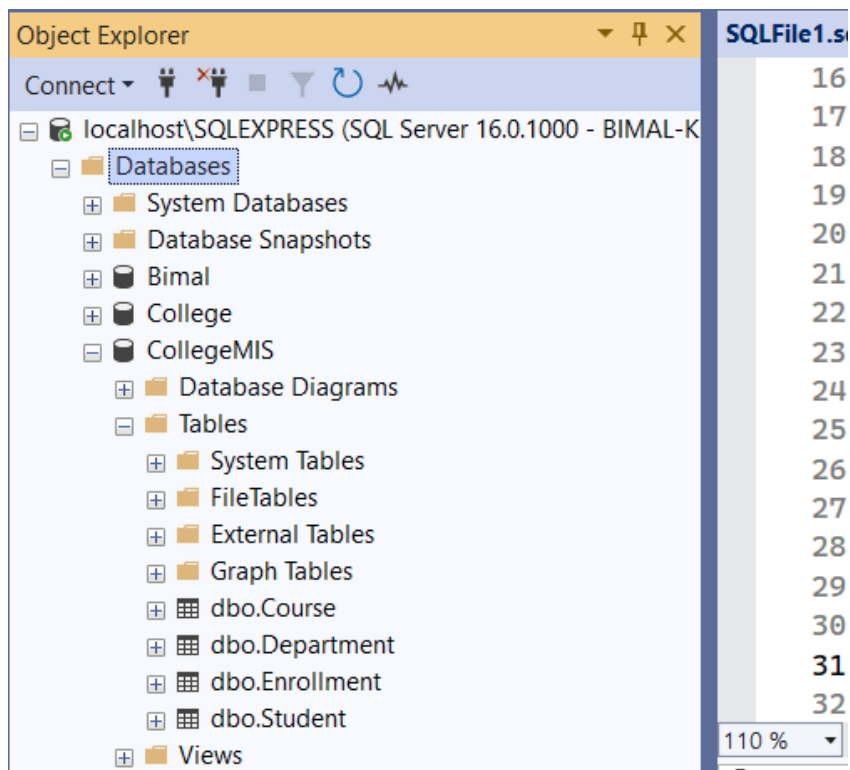
D. Creating Table of Course:

Query: CREATE TABLE Course (
 CourseID INT PRIMARY KEY,
 CourseName VARCHAR(100),
 DepartmentID INT,
 FOREIGN KEY (DepartmentID) REFERENCES
 Department(DepartmentID));



E. Creating Table of Enrollement:

Query: CREATE TABLE Enrollment (
 EnrollmentID INT PRIMARY KEY,
 StudentID INT,
 CourseID INT,
 Semester VARCHAR(10),
 Marks INT,
 FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
 FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);



1. Insert sample data into all four tables (Student, Department, Course, Enrollment).

Query:

```
INSERT INTO Student VALUES
```

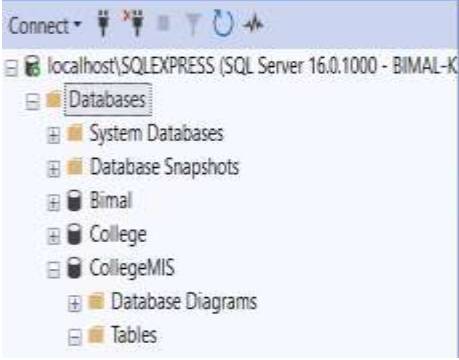
```
(101, 'Monica Adhikari', 'F', '2001-05-15', 1, 'monica@gmail.com'),
```

```
(102, 'Bimal Kunwar', 'M', '2000-08-20', 2, 'bimal@yahoo.com'),
```

```
(103, 'Laseeb Xettri', 'F', '2002-02-10', 1, 'laseeb@gmail.com'),
```

```
(104, 'Ritika Joshi', 'F', '2001-11-11', 3, 'ritika@outlook.com'),
```

```
(105, 'Bishal Shrestha', 'M', '2003-04-25', 2, 'bishal@gmail.com');
```



	StudentID	Name	Gender	DOB	DepartmentID	Email
1	101	Monica Adhikari	F	2001-05-15	1	monica@gmail.com
2	102	Bimal Kunwar	M	2000-08-20	2	bimal@yahoo.com
3	103	Laseeb Xettri	F	2002-02-10	1	laseeb@gmail.com
4	104	Ritika Joshi	F	2001-11-11	3	ritika@outlook.com
5	105	Bishal Shrestha	M	2003-04-25	2	bishal@gmail.com

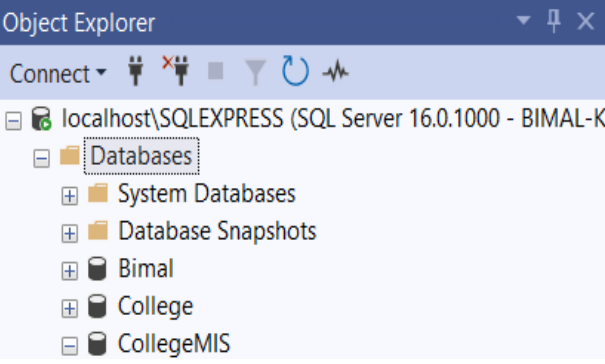
```
INSERT INTO Department (DepartmentID, DeptName) VALUES
```

```
(1, 'Computer Science'),
```

```
(2, 'Information Technology'),
```

```
(3, 'Electronics'),
```

```
(4, 'Mechanical');
```



	DepartmentID	DeptName
1	1	Computer Science
2	2	Information Technology
3	3	Electronics
4	4	Mechanical

INSERT INTO Course VALUES

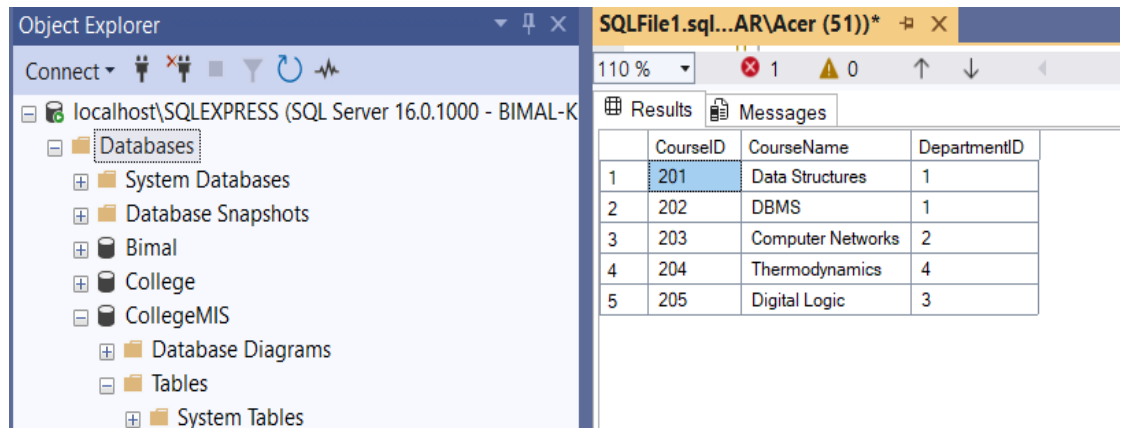
(201, 'Data Structures', 1),

(202, 'DBMS', 1),

(203, 'Computer Networks', 2),

(204, 'Thermodynamics', 4),

(205, 'Digital Logic', 3);



	CourseID	CourseName	DepartmentID
1	201	Data Structures	1
2	202	DBMS	1
3	203	Computer Networks	2
4	204	Thermodynamics	4
5	205	Digital Logic	3

INSERT INTO Enrollment VALUES

(301, 101, 201, 'Spring', 88),

(302, 102, 203, 'Spring', 76),

(303, 103, 201, 'Fall', 92),

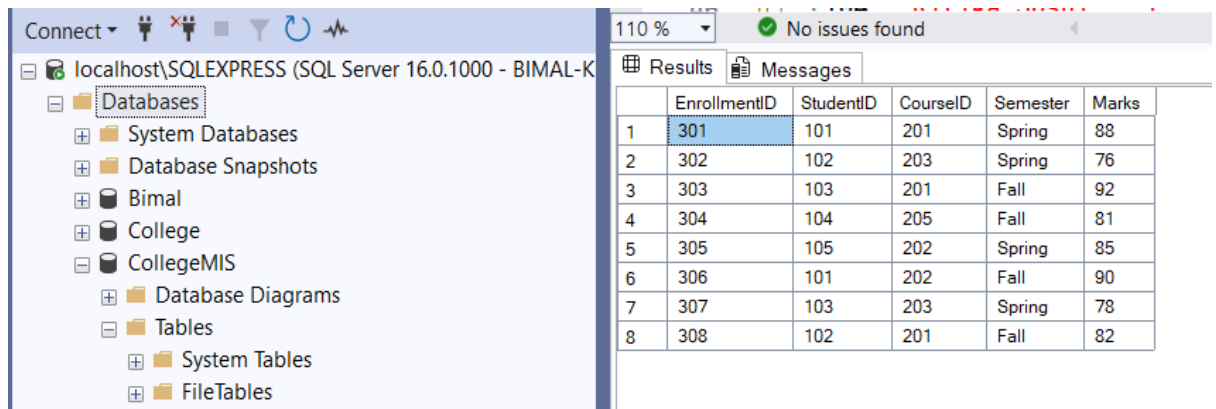
(304, 104, 205, 'Fall', 81),

(305, 105, 202, 'Spring', 85),

(306, 101, 202, 'Fall', 90),

(307, 103, 203, 'Spring', 78),

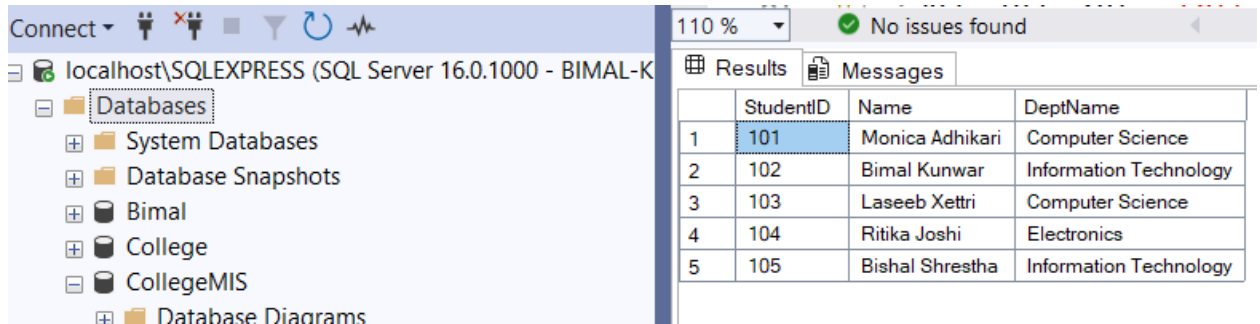
(308, 102, 201, 'Fall', 82);



	EnrollmentID	StudentID	CourseID	Semester	Marks
1	301	101	201	Spring	88
2	302	102	203	Spring	76
3	303	103	201	Fall	92
4	304	104	205	Fall	81
5	305	105	202	Spring	85
6	306	101	202	Fall	90
7	307	103	203	Spring	78
8	308	102	201	Fall	82

2. Display all students with their department name using JOIN.

```
Query: SELECT S.StudentID, S.Name, D.DeptName
FROM Student S
JOIN Department D ON S.DepartmentID = D.DepartmentID;
```

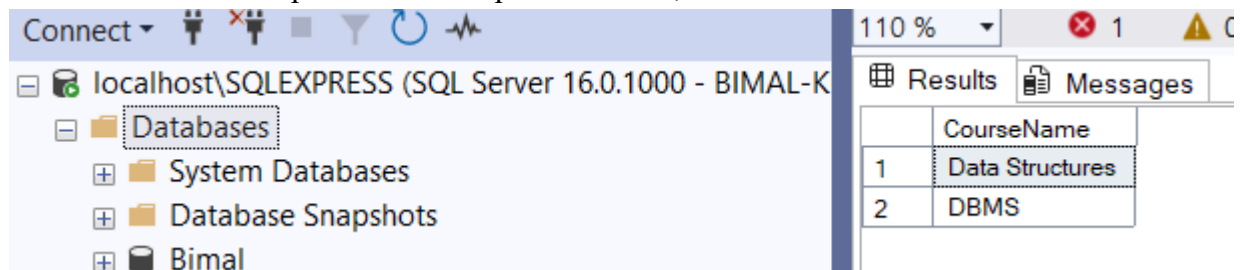


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded under 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. On the right, the 'Results' tab displays a table with 5 rows and 4 columns: StudentID, Name, and DeptName. The data is as follows:

	StudentID	Name	DeptName
1	101	Monica Adhikari	Computer Science
2	102	Bimal Kunwar	Information Technology
3	103	Laseeb Xettri	Computer Science
4	104	Ritika Joshi	Electronics
5	105	Bishal Shrestha	Information Technology

3. Show all courses offered by the "Computer Science" department.

```
Query: SELECT CourseName
FROM Course C
JOIN Department D ON C.DepartmentID = D.DepartmentID
WHERE D.DeptName = 'Computer Science';
```

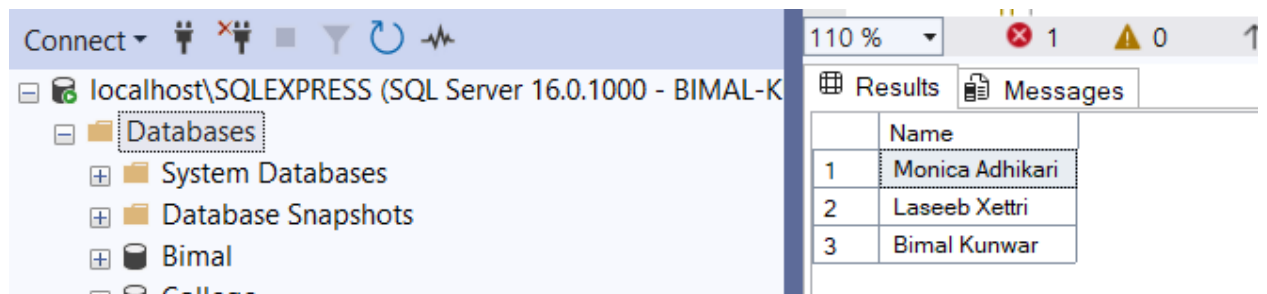


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded under 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. On the right, the 'Results' tab displays a table with 2 rows and 2 columns: CourseName. The data is as follows:

	CourseName
1	Data Structures
2	DBMS

4. List all students enrolled in "Data Structures" course.

```
Query: SELECT DISTINCT S.Name
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
JOIN Course C ON E.CourseID = C.CourseID
WHERE C.CourseName = 'Data Structures';
```

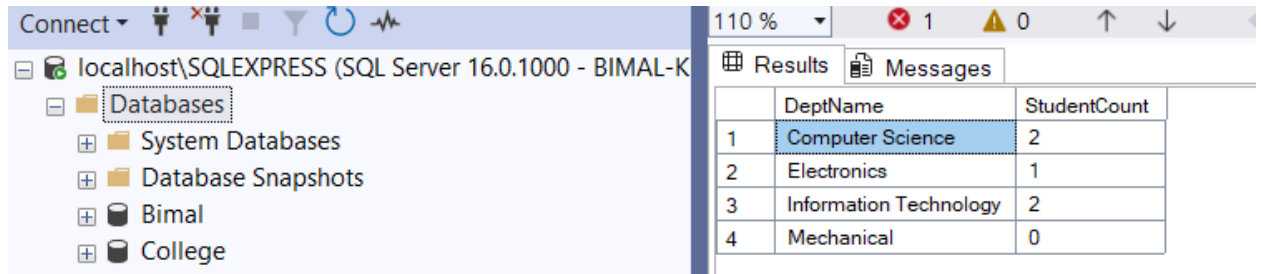


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded under 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. On the right, the 'Results' tab displays a table with 3 rows and 2 columns: Name. The data is as follows:

	Name
1	Monica Adhikari
2	Laseeb Xettri
3	Bimal Kunwar

5. Count the number of students in each department.

```
Query: SELECT D.DeptName, COUNT(S.StudentID) AS StudentCount
FROM Department D
LEFT JOIN Student S ON D.DepartmentID = S.DepartmentID
GROUP BY D.DeptName;
```

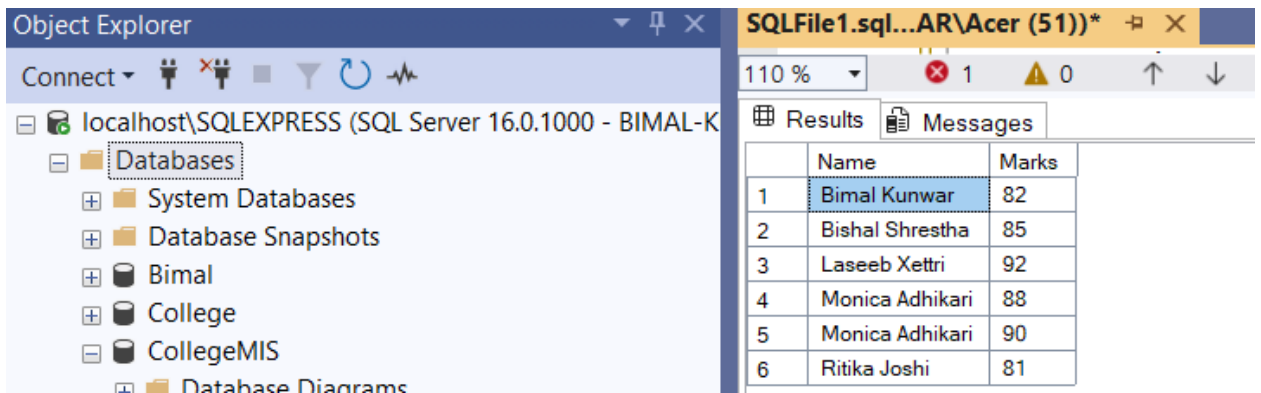


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded under 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. On the right, the 'Results' tab is active, displaying a table with two columns: 'DeptName' and 'StudentCount'. The table contains four rows of data.

	DeptName	StudentCount
1	Computer Science	2
2	Electronics	1
3	Information Technology	2
4	Mechanical	0

6. List students who scored more than 80 in any course.

```
Query: SELECT DISTINCT S.Name, E.Marks
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
WHERE E.Marks > 80;
```

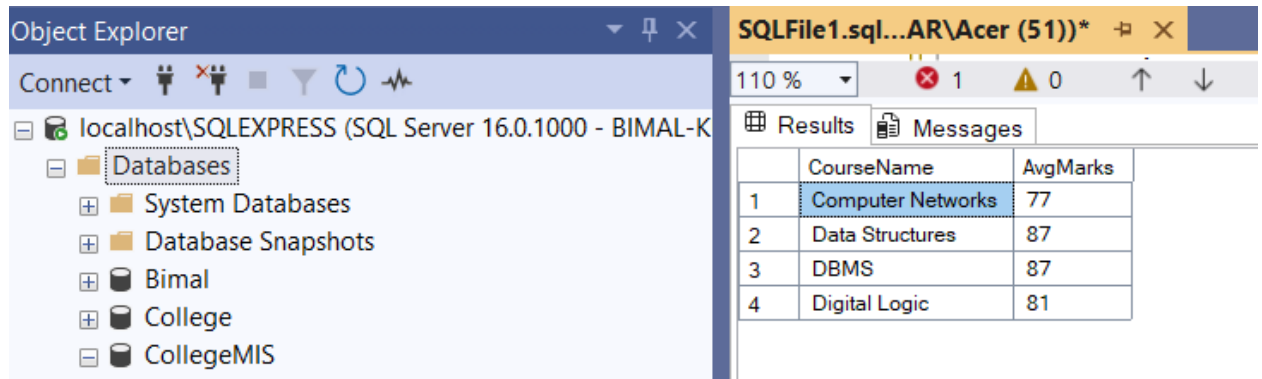


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded under 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. On the right, the 'Results' tab is active, displaying a table with two columns: 'Name' and 'Marks'. The table contains six rows of data.

	Name	Marks
1	Bimal Kunwar	82
2	Bishal Shrestha	85
3	Laseeb Xettri	92
4	Monica Adhikari	88
5	Monica Adhikari	90
6	Ritika Joshi	81

7. Find the average marks for each course using GROUP BY.

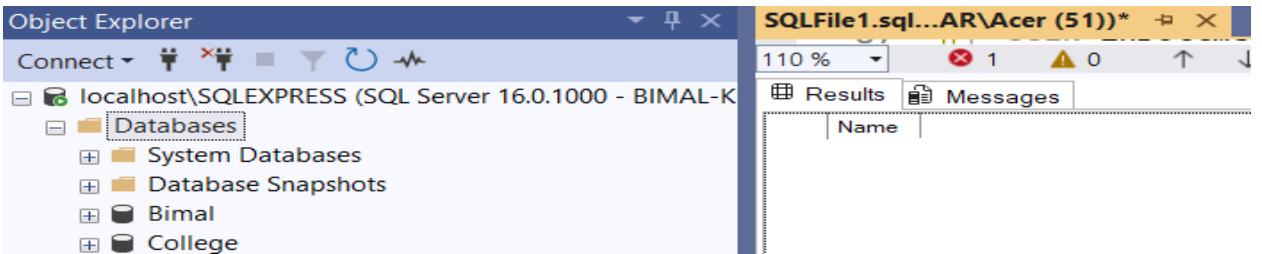
```
Query: SELECT C.CourseName, AVG(E.Marks) AS AvgMarks
        FROM Course C
        JOIN Enrollment E ON C.CourseID = E.CourseID
        GROUP BY C.CourseName;
```



	CourseName	AvgMarks
1	Computer Networks	77
2	Data Structures	87
3	DBMS	87
4	Digital Logic	81

8. Display students who have not enrolled in any course.

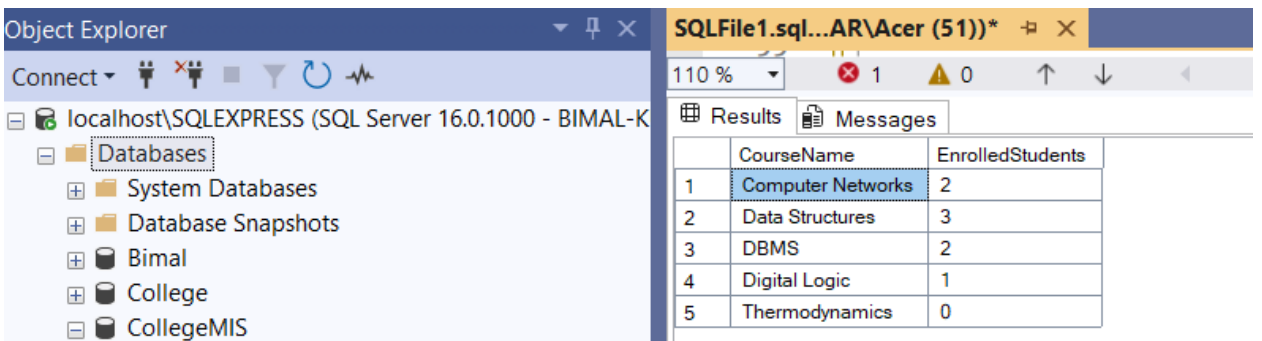
```
Query: SELECT S.Name
        FROM Student S
        LEFT JOIN Enrollment E ON S.StudentID = E.StudentID
        WHERE E.EnrollmentID IS NULL;
```



Name

9. List all courses with the number of students enrolled in them.

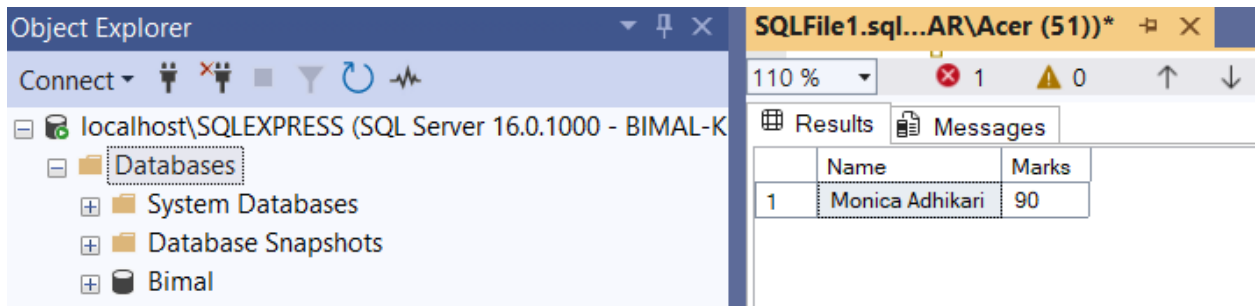
```
Query: SELECT C.CourseName, COUNT(E.StudentID) AS EnrolledStudents
        FROM Course C
        LEFT JOIN Enrollment E ON C.CourseID = E.CourseID
        GROUP BY C.CourseName;
```



	CourseName	EnrolledStudents
1	Computer Networks	2
2	Data Structures	3
3	DBMS	2
4	Digital Logic	1
5	Thermodynamics	0

10. Find the student with the highest marks in "DBMS" course.

```
Query: SELECT S.StudentID, S.Name, E.Marks
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
JOIN Course C ON E.CourseID = C.CourseID
WHERE C.CourseName = 'DBMS'
AND E.Marks = (
    SELECT MAX(E2.Marks)
    FROM Enrollment E2
JOIN Course C2 ON E2.CourseID = C2.CourseID
WHERE C2.CourseName = 'DBMS'
);
```

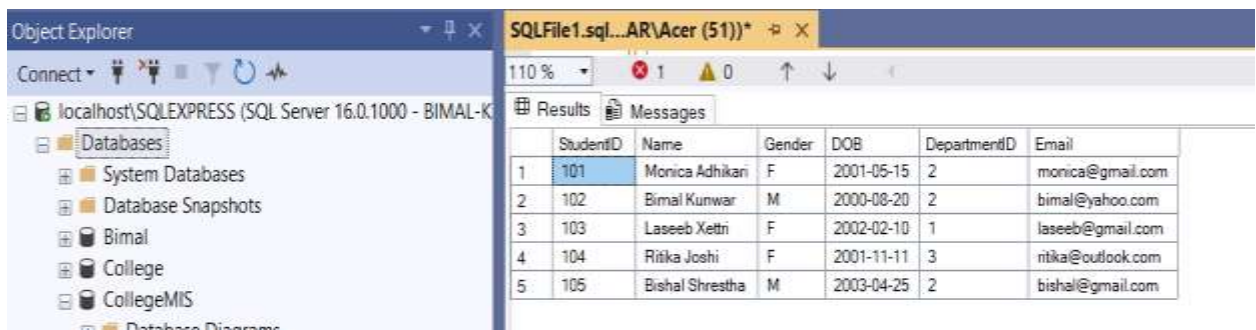


The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The main window shows the results of the query in a table with two columns: 'Name' and 'Marks'. The result shows a single row for 'Monica Adhikari' with a mark of 90.

	Name	Marks
1	Monica Adhikari	90

11. Update a student's department based on StudentID.

```
Query: UPDATE Student
SET DepartmentID = 2
WHERE StudentID = 101;
```

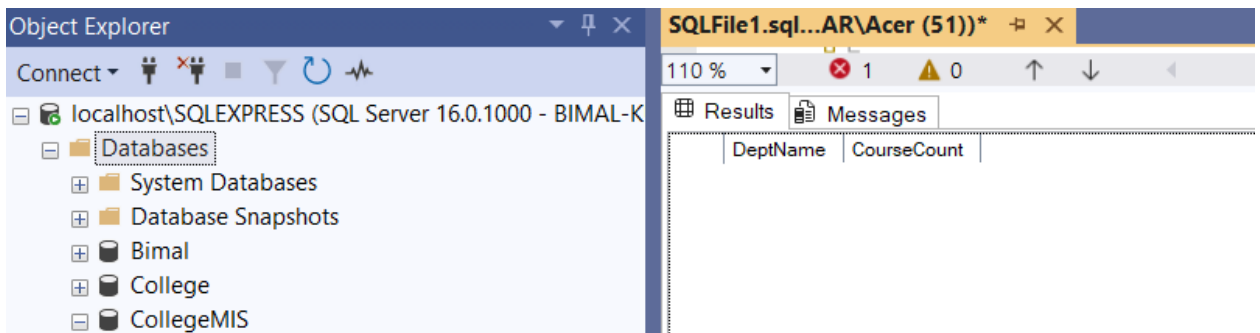


The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The main window shows the results of the update operation in a table with seven columns: 'StudentID', 'Name', 'Gender', 'DOB', 'DepartmentID', and 'Email'. The result shows five rows of student data.

	StudentID	Name	Gender	DOB	DepartmentID	Email
1	101	Monica Adhikari	F	2001-05-15	2	monica@gmail.com
2	102	Bimal Kunwar	M	2000-08-20	2	bimal@yahoo.com
3	103	Laseeb Xettri	F	2002-02-10	1	laseeb@gmail.com
4	104	Ritika Joshi	F	2001-11-11	3	ritika@outlook.com
5	105	Bishal Shrestha	M	2003-04-25	2	bishal@gmail.com

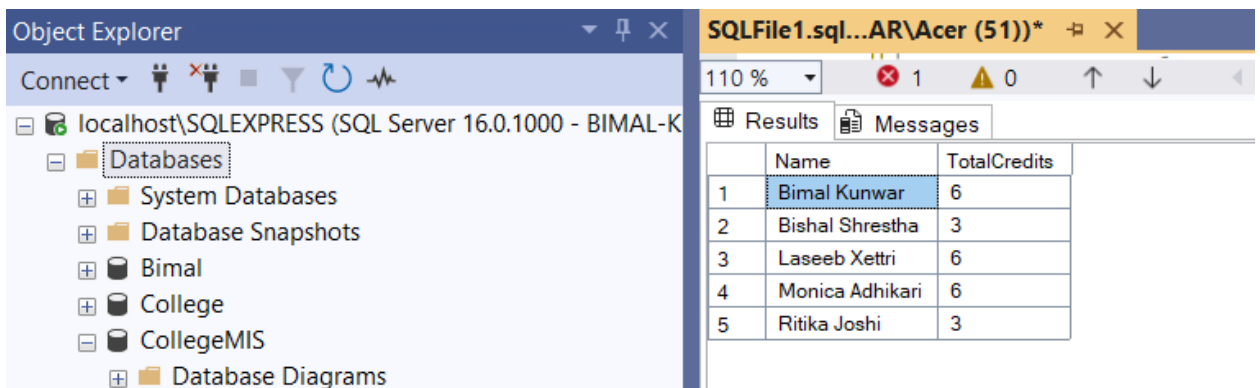
12. Find the departments that offer more than 3 courses.

```
Query: SELECT D.DeptName, COUNT(C.CourseID) AS CourseCount
FROM Department D
JOIN Course C ON D.DepartmentID = C.DepartmentID
GROUP BY D.DeptName
HAVING COUNT(C.CourseID) > 3;
```



13. Show the names of students along with total credits they are enrolled in.

```
Query: SELECT S.Name, COUNT(E.CourseID) AS TotalCourses
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
GROUP BY S.Name;
```



14. Create a view TopPerformers that shows students with average marks > 85.

```
Query: CREATE VIEW TopPerformers AS
SELECT S.StudentID, S.Name, AVG(E.Marks) AS AvgMarks
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
GROUP BY S.StudentID, S.Name
HAVING AVG(E.Marks) > 85;
```

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The 'Views' folder is expanded, and 'dbo.TopPerformers' is highlighted. On the right, the SQL File1 window shows the query results for the 'TopPerformers' view. The results are displayed in a table with columns: StudentID, Name, and AvgMarks. The table contains one row: StudentID 101, Name Monica Adhikari, and AvgMarks 89.

	StudentID	Name	AvgMarks
1	101	Monica Adhikari	89

15. Create a view StudentDetails to show StudentID, Name, Department, and Email

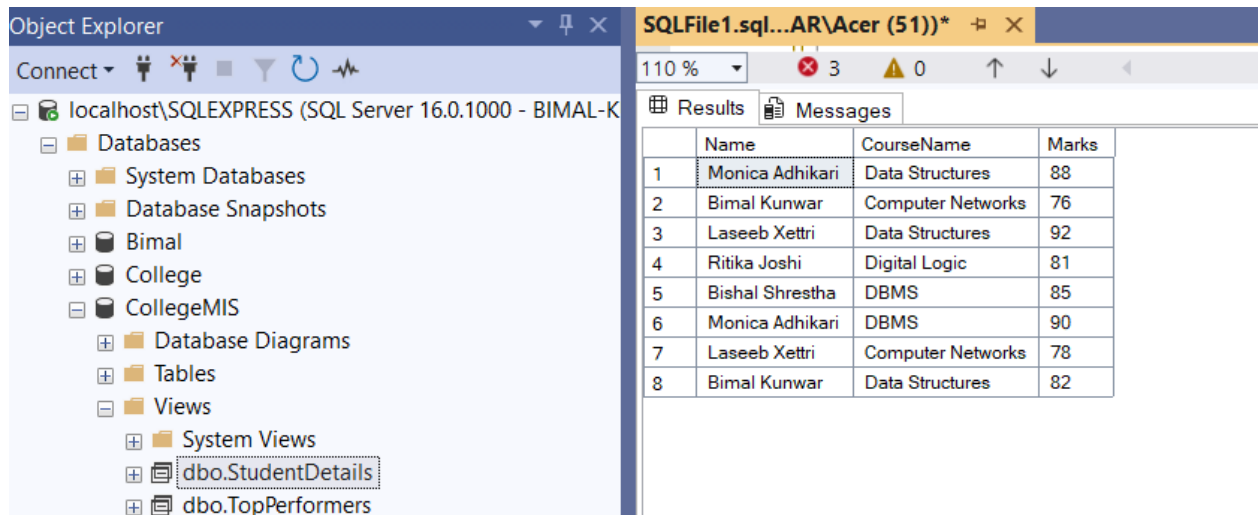
```
Query: CREATE VIEW StudentDetails AS
SELECT S.StudentID, S.Name, D.DeptName AS Department, S.Email
FROM Student S
JOIN Department D ON S.DepartmentID = D.DepartmentID;
```

The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The 'Views' folder is expanded, and 'dbo.StudentDetails' is highlighted. On the right, the SQL File1 window shows the query results for the 'StudentDetails' view. The results are displayed in a table with columns: StudentID, Name, DeptName, and Email. The table contains five rows of student data.

	StudentID	Name	DeptName	Email
1	101	Monica Adhikari	Information Technology	monica@gmail.com
2	102	Bimal Kunwar	Information Technology	bimal@yahoo.com
3	103	Laseeb Xettri	Computer Science	laseeb@gmail.com
4	104	Ritika Joshi	Electronics	ritika@outlook.com
5	105	Bishal Shrestha	Information Technology	bishal@gmail.com

16. Write a query to show student name, course name, and marks for all enrollments

```
Query: SELECT S.Name AS StudentName, C.CourseName, E.Marks
FROM Enrollment E
JOIN Student S ON E.StudentID = S.StudentID
JOIN Course C ON E.CourseID = C.CourseID;
```

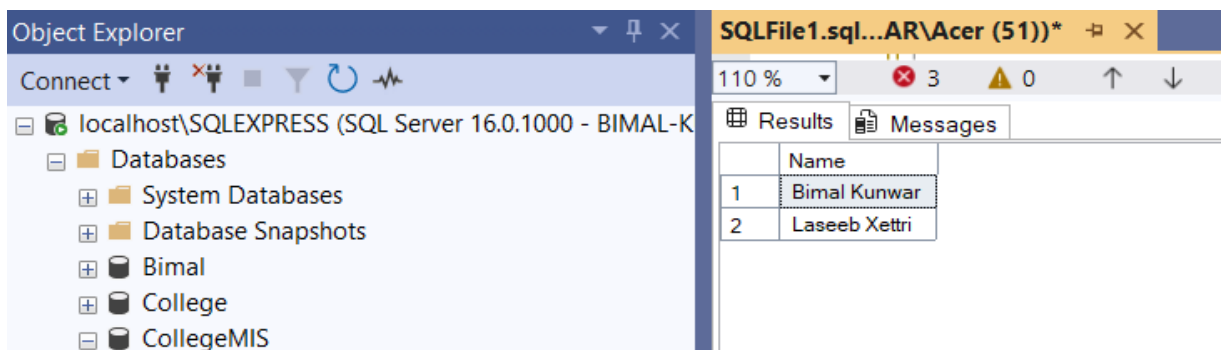


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The 'Databases' folder is expanded, showing 'System Databases', 'Database Snapshots', 'Bimal', 'College', and 'CollegeMIS'. Under 'CollegeMIS', 'Database Diagrams', 'Tables', and 'Views' are listed. The 'Views' folder is expanded, showing 'System Views', 'dbo.StudentDetails', and 'dbo.TopPerformers'. The 'dbo.StudentDetails' view is selected. On the right, the 'Results' pane shows the output of a query. The query is 'SELECT S.Name AS StudentName, C.CourseName, E.Marks FROM Enrollment E JOIN Student S ON E.StudentID = S.StudentID JOIN Course C ON E.CourseID = C.CourseID;'. The results are displayed in a table with 8 rows and 3 columns: 'Name', 'CourseName', and 'Marks'.

	Name	CourseName	Marks
1	Monica Adhikari	Data Structures	88
2	Bimal Kunwar	Computer Networks	76
3	Laseeb Xettri	Data Structures	92
4	Ritika Joshi	Digital Logic	81
5	Bishal Shrestha	DBMS	85
6	Monica Adhikari	DBMS	90
7	Laseeb Xettri	Computer Networks	78
8	Bimal Kunwar	Data Structures	82

17. Find students who have enrolled in courses from more than one department.

```
Query: SELECT S.Name
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
JOIN Course C ON E.CourseID = C.CourseID
GROUP BY S.StudentID, S.Name
HAVING COUNT(DISTINCT C.DepartmentID) > 1;
```

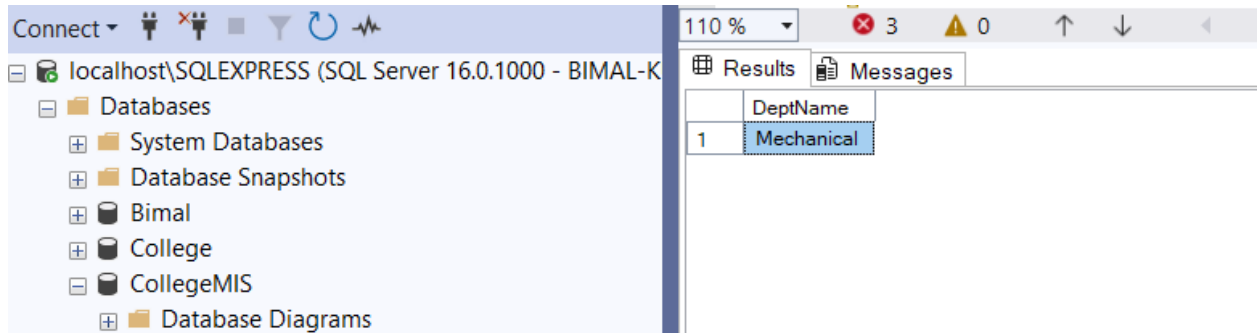


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the Object Explorer displays the database structure for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The 'Databases' folder is expanded, showing 'System Databases', 'Database Snapshots', 'Bimal', 'College', and 'CollegeMIS'. Under 'CollegeMIS', 'Database Diagrams', 'Tables', and 'Views' are listed. The 'Views' folder is expanded, showing 'System Views', 'dbo.StudentDetails', and 'dbo.TopPerformers'. The 'dbo.StudentDetails' view is selected. On the right, the 'Results' pane shows the output of a query. The query is 'SELECT S.Name FROM Student S JOIN Enrollment E ON S.StudentID = E.StudentID JOIN Course C ON E.CourseID = C.CourseID GROUP BY S.StudentID, S.Name HAVING COUNT(DISTINCT C.DepartmentID) > 1;'. The results are displayed in a table with 2 rows and 1 column: 'Name'.

	Name
1	Bimal Kunwar
2	Laseeb Xettri

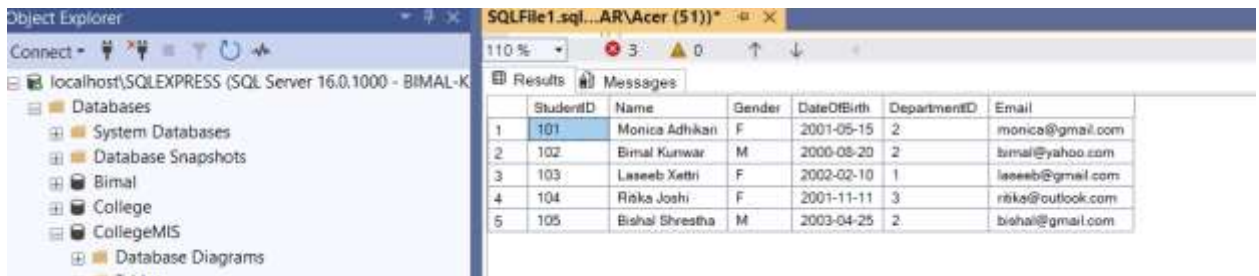
18. Display the list of departments that currently have no students

```
Query: SELECT DeptName
FROM Department D
LEFT JOIN Student S ON D.DepartmentID = S.DepartmentID
WHERE S.StudentID IS NULL;
```



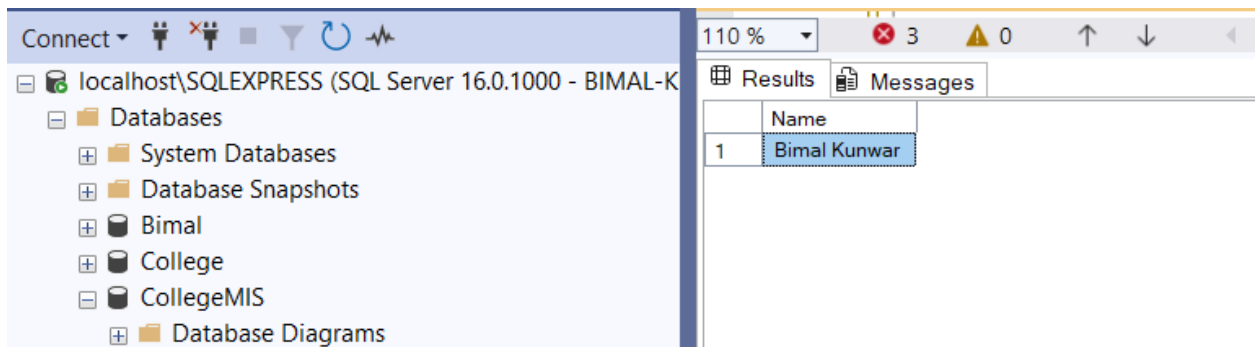
19. Rename the column DOB in Student to DateOfBirth.

```
Query: EXEC sp_rename 'Student.DOB', 'DateOfBirth', 'COLUMN';
```



20. Find students who are both in the "IT" department and enrolled in "Computer Networks"

```
Query: SELECT S.Name
        FROM Student S
        JOIN Enrollment E ON S.StudentID = E.StudentID
        JOIN Course C ON E.CourseID = C.CourseID
        JOIN Department D ON S.DepartmentID = D.DepartmentID
        WHERE D.DeptName = 'IT' AND C.CourseName = 'Computer Networks';
```

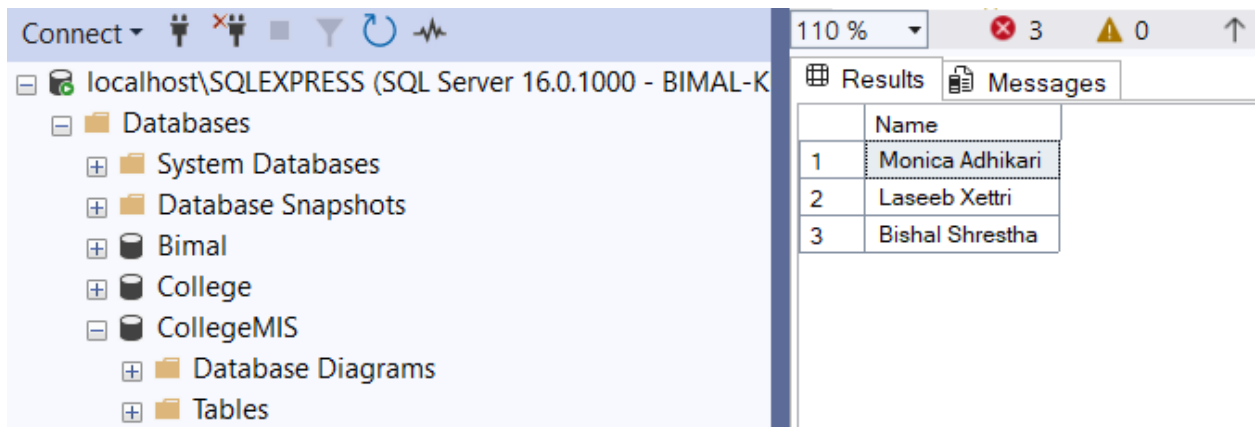


The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the server hierarchy for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The right pane shows the 'Results' tab with a single row of data.

	Name
1	Bimal Kunwar

21. Find students whose email ends with '@gmail.com'

```
Query: SELECT Name, Email
        FROM Student
        WHERE Email LIKE '%@gmail.com';
```

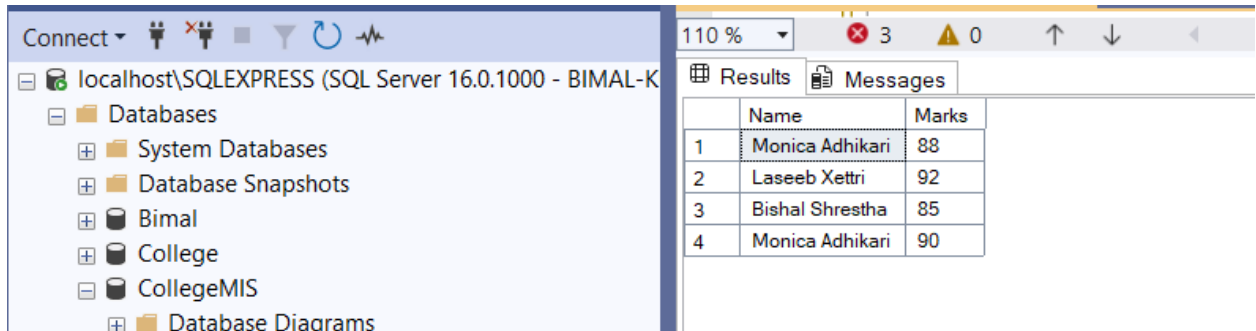


The screenshot shows the SQL Server Enterprise Manager interface. The left pane displays the server hierarchy for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The right pane shows the 'Results' tab with three rows of data.

	Name
1	Monica Adhikari
2	Laseeb Xettri
3	Bishal Shrestha

22. Display students who scored more than the average mark of all students.

```
Query: SELECT S.Name, E.Marks
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
WHERE E.Marks > (SELECT AVG(Marks) FROM Enrollment);
```

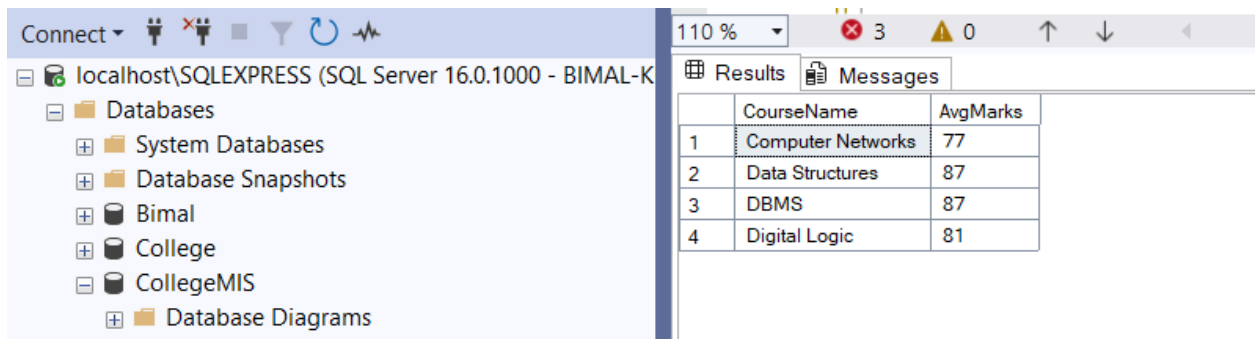


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded, showing 'System Databases', 'Database Snapshots', 'Bimal', 'College', 'CollegeMIS', and 'Database Diagrams'. On the right, the 'Results' pane displays the output of the query. The table has two columns: 'Name' and 'Marks'. The results are as follows:

	Name	Marks
1	Monica Adhikari	88
2	Laseeb Xettri	92
3	Bishal Shrestha	85
4	Monica Adhikari	90

23. Show the average marks of students in each course.

```
Query: SELECT C.CourseName, AVG(E.Marks) AS AvgMarks
FROM Course C
JOIN Enrollment E ON C.CourseID = E.CourseID
GROUP BY C.CourseName;
```

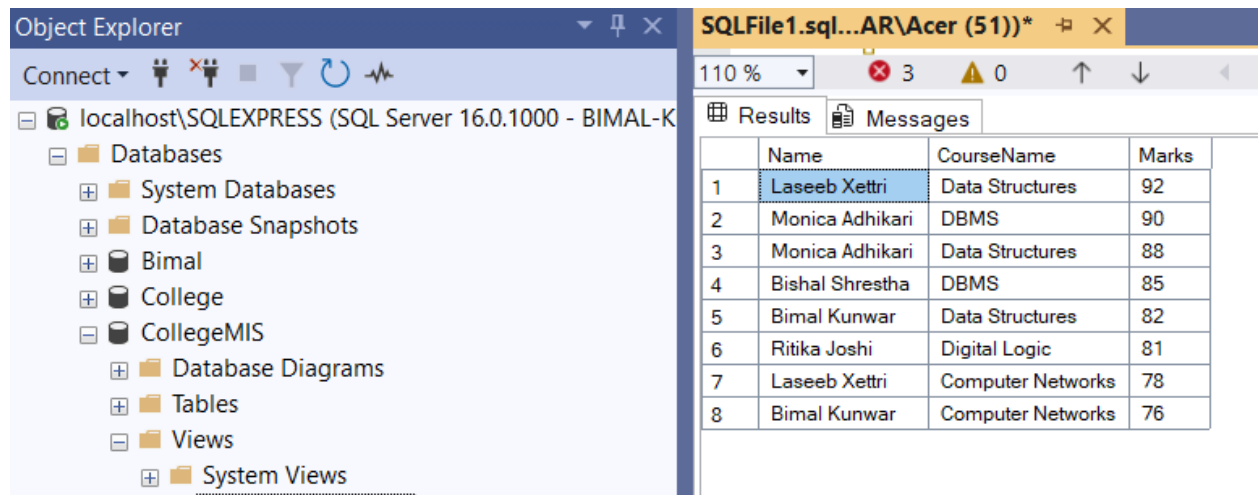


The screenshot shows the SQL Server Enterprise Manager interface. On the left, the 'Databases' folder is expanded, showing 'System Databases', 'Database Snapshots', 'Bimal', 'College', 'CollegeMIS', and 'Database Diagrams'. On the right, the 'Results' pane displays the output of the query. The table has two columns: 'CourseName' and 'AvgMarks'. The results are as follows:

	CourseName	AvgMarks
1	Computer Networks	77
2	Data Structures	87
3	DBMS	87
4	Digital Logic	81

24. Show all students with course name and sort by Marks descending.

```
Query: SELECT S.Name AS StudentName, C.CourseName, E.Marks
FROM Student S
JOIN Enrollment E ON S.StudentID = E.StudentID
JOIN Course C ON E.CourseID = C.CourseID
ORDER BY E.Marks DESC;
```



The screenshot shows the SQL Server Enterprise Manager interface. The Object Explorer on the left displays the database structure for 'localhost\SQLEXPRESS (SQL Server 16.0.1000 - BIMAL-K)'. The Results pane on the right shows the output of the query, displaying a table with 8 rows of student and course data, sorted by marks in descending order.

	Name	CourseName	Marks
1	Laseeb Xettri	Data Structures	92
2	Monica Adhikari	DBMS	90
3	Monica Adhikari	Data Structures	88
4	Bishal Shrestha	DBMS	85
5	Bimal Kunwar	Data Structures	82
6	Ritika Joshi	Digital Logic	81
7	Laseeb Xettri	Computer Networks	78
8	Bimal Kunwar	Computer Networks	76